

Gemini CLI 介紹與上手

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<https://bit.ly/sakana20251130>

About me

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- Google Cloud GDE
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Gemini CLI 介紹

Gemini CLI : 你的開源 AI 終端機助理

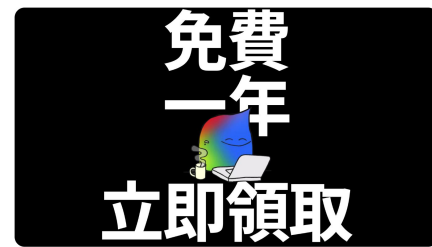
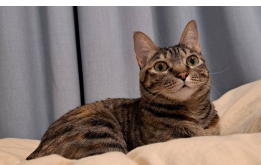
- 官網 <https://geminicli.com/>
- Github <https://github.com/google-gemini/gemini-cli>
- Google 文件 <https://cloud.google.com/gemini/docs/codeassist/gemini-cli?hl=zh-tw>
- 特色
 - 一個互動式的命令列介面 (CLI) 工具, 使用 Gemini 語言模型。
 - 不只是聊天, 能實際操作本機檔案系統、執行指令、搜尋網路。
 - 開源且具備高擴充性 (支援 MCP 或擴充功能支援)



Google AI Pro 大學生免費方案

官方頁面: <https://gemini.google/students/>

- 18 歲以上大學生 / 2025/12/9 申請完成
- 官方說明: <https://support.google.com/gemini/answer/16417758?hl=zh-Hant>
- Google AI Pro 福利: <https://support.google.com/googleone/answer/14534406>
- 網路搜尋文章 [瘋先生]
 - <https://mrmad.com.tw/taiwan-google-ai-pro-student-free-year-guide>



Gemini CLI 取得 - GCP Cloud Shell

GCP 專案開啟 Cloud Shell

- 內建 Gemini CLI
- 同時有 OSS code 可使用
- Cloud Shell
 - 每月有 60 小時可以使用,
4 vCPU / 16 GB MEM

A screenshot of the GCP Cloud Shell Editor interface. The browser tab is titled 'Cloud Shell Editor'. The interface shows a terminal window with the 'GEMINI' logo in large, colorful, pixelated letters. Below the logo, there are 'Tips for getting started' and a message about running Gemini CLI in a project-specific directory. At the bottom, there is a prompt line '> █ Type your message or @path/to/file' and status information: 'no sandbox (see /docs)' and 'gemini-2.5-pro (100% context left)'.

```
Cloud Shell Editor

cloudshell x Gemini CLI x +

GEMINI

Tips for getting started:
1. Ask questions, edit files, or run commands.
2. Be specific for the best results.
3. Create GEMINI.md files to customize your interactions with Gemini.
4. /help for more information.

You are running Gemini CLI in your home directory. It is recommended to
run in a project-specific directory.

> █ Type your message or @path/to/file

~

no sandbox (see /docs)

gemini-2.5-pro (100% context left)
```

Gemini CLI 取得 - 自行安裝

可參考Github 網頁

- <https://github.com/google-gemini/gemini-cli?tab=readme-ov-file#quick-install>
 - 建議可使用 `npm install -g @google/gemini-cli` 方式安裝
 - Node.js 版本 20 以上
 - node.js 安裝可參考 <https://nodejs.org/en/download>
 - 請注意部分 linux 預設 node.js 版本為 18
 - 可以參考這篇文章調整
<https://sakananote2.blogspot.com/2025/10/ubuntu-2404-lts-nvm-nodejs.html>



啟動 Gemini CLI

啟動警語

- "投資有賺有賠, AI 也有**幻覺**, 請詳閱**輸出入訊息**"

直接輸入 gemini 啟動

- 啟動目錄為工作目錄
- 可用 `--include-directories` 增加額外的工作目錄
- `/directory` show 與 add
- `/settings` Preview Features

> 你是不是貼錯了

+ 您是對的，非常抱歉，我複製錯了 `new_string` 的內容，導致上一個更新指令是無效的。我會立即修正這個錯誤。
我現在重新讀取計畫檔案，並使用正確的內容來更新第三步的狀態。

✓ ReadFile PROJECT_PLAN.md

+ 好的，我現在使用正確的內容來更新計畫檔案。

```
max@Max-MacAir temp % gemini
Loaded cached credentials.

GEMINI

Tips for getting started:
1. Ask questions, edit files, or run commands.
2. Be specific for the best results.
3. /help for more information.

> 請幫我查看台北今天天氣 將輸出建立一個檔案到 /Users/max/test.txt

✓ GoogleSearch Searching the web for: "台北今天天氣"
Search results for "台北今天天氣" returned.

✗ WriteFile {"file_path":"/Users/max/test.txt","content":"臺北今天的天氣多...

File path must be within one of the workspace
directories: /Users/max/Downloads/local_lab/temp
```



Gemini CLI - 驗證方式

Gemini CLI 有 3 種驗證方式 `/auth` , 使用 `/about` 觀察

- 可參考文章

<https://sakananote2.blogspot.com/2025/09/gemini-cli-login.html>

- Login with Google

- 個人付費 / 免費用戶 (60 requests/min and 1,000 requests/day)

- Use Gemini API Key

- Developers who need specific model control

- Vertex AI

Enterprise teams and production workloads



Gemini CLI - 觀察功能

輸入 `/help` 觀察

- `@` 指定檔案或目錄 (或要求他先讀取指定檔案)
- `/quit` 離開 Gemini CLI
 - 或是 2 次 Ctrl + C
- `/theme` 設定顏色與主題
- `/tools` 觀察功能
- 使用 `!` 符號可以執行終端機指令 (例如 `!ls -l`), 使用 Ctrl + F 切換焦點. Ctrl + S 啟用 選擇模式

```
Basic:
Add context: Use @ to specify files for context (e.g., @src/myFile.ts) to target specific files or folders.
Shell mode: Execute shell commands via ! (e.g., !npm run start) or use natural language (e.g. start server).

Commands:
/about - show version info
/auth - change the auth method
/bug - submit a bug report
/chat - Manage conversation history.
  list - List saved conversation checkpoints
  save - Save the current conversation as a checkpoint. Usage: /chat save <tag>
  resume - Resume a conversation from a checkpoint. Usage: /chat resume <tag>
  delete - Delete a conversation checkpoint. Usage: /chat delete <tag>
  share - Share the current conversation to a markdown or json file. Usage: /chat share <file>
/clear - clear the screen and conversation history
/compress - Compresses the context by replacing it with a summary.
/copy - Copy the last result or code snippet to clipboard
/docs - open full Gemini CLI documentation in your browser
/directory - Manage workspace directories
  add - Add directories to the workspace. Use comma to separate multiple paths
  show - Show all directories in the workspace
/editor - set external editor preference
/extensions - Manage extensions
  list - List active extensions
  update - Update extensions. Usage: update <extension-names>|--all
/help - for help on gemini-cli
/ide - manage IDE integration
/init - Analyzes the project and creates a tailored GEMINI.md file.
/mcp - list configured MCP servers and tools, or authenticate with OAuth-enabled servers
  list - List configured MCP servers and tools
  auth - Authenticate with an OAuth-enabled MCP server
  refresh - Restarts MCP servers.
/memory - Commands for interacting with memory.
  show - Show the current memory contents.
  add - Add content to the memory.
  refresh - Refresh the memory from the source.
/privacy - display the privacy notice
/quit - exit the cli
/stats - check session stats. Usage: /stats [model][tools]
  model - Show model-specific usage statistics.
  tools - Show tool-specific usage statistics.
/theme - change the theme
/tools - list available Gemini CLI tools. Usage: /tools [desc]
/settings - View and edit Gemini CLI settings
/vim - toggle vim mode on/off
/setup-github - Set up GitHub Actions
/terminal-setup - Configure terminal keybindings for multiline input (VS Code, Cursor, Windsurf)
! - shell command
[MCP] - Model Context Protocol command (from external servers)
```

```
> /tools
```

Available Gemini CLI tools:

- Edit
- FindFiles
- GoogleSearch
- ReadFile
- ReadFolder
- ReadManyFiles
- Save Memory
- SearchText
- Shell
- WebFetch
- WriteFile



Gemini CLI - 設定檔

官方文件

- <https://geminicli.com/docs/get-started/configuration/>
- settings.json
 - 驗證方式 / 主題 / MCP ...
 - workspace: .gemini/settings.json
 - 使用者: ~/.gemini/settings.json
- GEMINI.md
 - 自訂 Gemini CLI
 - ~/.gemini/GEMINI.md / 專案根目錄 / 子目錄
 - 使用 /memory show 觀察



Gemini CLI - options

Gemini CLI options

- -m --model 選模型
 - 目前 預設是使用 auto, 可利用 /model 指定
- -p --prompt 非互動方式
 - 可參考 [Headless Mode](#)
 - % echo "介紹虎斑貓" | gemini
 - gemini -p "介紹虎斑貓"
 - 可與 cron job 結合, 一般來說會搭配 --yolo
- -y --yolo (You Only Live Once) Automatically accept all actions



Gemini CLI - MCP

Model Context Protocol (MCP)

- 建立一套標準化的規範，讓人工智慧系統，特別是大型語言模型 (LLMs)，能夠安全且高效地整合和存取外部工具、系統和數據。
- 參考文章 <https://sakananote2.blogspot.com/2025/10/chrome-devtools-mcp-with-gemini-cli.html>
 - `% gemini mcp add chrome-devtools npx chrome-devtools-mcp@latest`
 - "Check the performance of <https://developers.chrome.com> "
- Gemini CLI Extensions
 - <https://geminicli.com/extensions/>
 - <https://github.com/gemini-cli-extensions>
 - `gemini extension install <source>`
 - `gemini extension uninstall <name>`



Gemini CLI - vscode 整合

官方文件

- <https://geminicli.com/docs/ide-integration/> - 目前支援 vscode
 - 輸入 `/ide install` 安裝
 - 輸入 `/ide status` 觀察
- **檔案感知**: Gemini CLI 知道您目前在 IDE 中開啟了哪些檔案。
- **選取程式碼感知**: 當您在編輯器中選取一段程式碼時, CLI 中的提示 (Prompt) 會自動將這段程式碼納入情境。
- **原生差異檢視**: 當您要求 Gemini CLI 進程式碼修改時, 它會將建議的變更以 IDE 原生的差異檢視 (Diff Editor) 形式打開。



Gemini CLI - 練習

練習方式

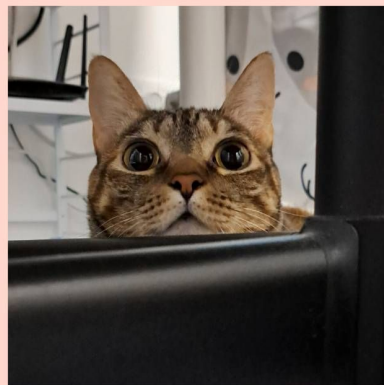
- 與 Gemini CLI 對話
- Codelabs 練習
 - <https://codelabs.developers.google.com/gemini-cli-hands-on?hl=zh-tw#0>
 - <https://codelabs.developers.google.com/agentverse-developer/instructions?hl=zh-tw#0>
- 咒語詠唱？
 - 請幫我撰寫一組用於 Google Cloud Platform (GCP) 的 ansible 設定檔，目標是建立一個 GCS 儲存桶。
 - 請幫我寫一個 Terraform 設定檔，用來在 Google Cloud Platform (GCP) 上建立一個標準儲存等級、位置在 asia-east1 的 GCS 儲存桶，Project 名稱與 GCS 名稱單獨儲存為變數，建立如何使用的中文 README.md 檔案，所有的檔案都要有中文註解。



無情工商

歡迎訂閱 :)

- YouTube
 - <https://youtube.com/@soyines1211>
- IG
 - @chenmicat
- TikTok
 - @ermi61474

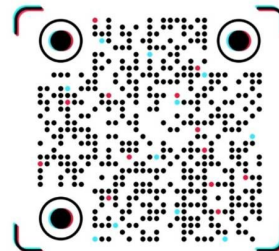


@CHENMICAT



二咪ermi

@ermi61474



TikTok

THANKS